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Inventor(s)	Liu; Chien-Heng et al.

Semiconductor device with a single diffusion break structure and a gate structure having aligned sidewalls

Abstract

A method for fabricating minimal fin length includes the steps of first forming a fin-shaped structure extending along a first direction on a substrate, forming a first single-diffusion break (SDB) trench and a second SDB trench extending along a second direction to divide the fin-shaped structure into a first portion, a second portion, and a third portion, and then performing a fin-cut process to remove the first portion and the third portion.

Inventors: Liu; Chien-Heng (Pingtung County, TW), Huang; Chia-Wei (Kaohsiung, TW), Yu; Hsin-Jen (Tainan, TW), Cheng; Yung-Feng (Kaohsiung, TW), Chen; Ming-Jui (Hsinchu, TW)

Applicant: UNITED MICROELECTRONICS CORP. (Hsin-Chu, TW)

Family ID: 81547044

Assignee: UNITED MICROELECTRONICS CORP. (Hsin-Chu, TW)

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation application of U.S. application Ser. No. 17/118,630, filed on Dec. 11, 2020. The content of the application is incorporated herein by reference.

BACKGROUND OF THE INVENTION

1. Field of the Invention

(1) The invention relates to a method for fabricating semiconductor device, and more particularly to a method of using single diffusion break (SDB) structure to enlarge photolithography window.

2. Description of the Prior Art

(2) With the trend in the industry being towards scaling down the size of the metal oxide semiconductor transistors (MOS), three-dimensional or non-planar transistor technology, such as fin field effect transistor technology (FinFET) has been developed to replace planar MOS transistors. Since the three-dimensional structure of a FinFET increases the overlapping area between the gate and the fin-shaped structure of the silicon substrate, the channel region can therefore be more effectively controlled. This way, the drain-induced barrier lowering (DIBL) effect and the short channel effect are reduced. The channel region is also longer for an equivalent gate length, thus the current between the source and the drain is increased. In addition, the threshold voltage of the fin FET can be controlled by adjusting the work function of the gate.

(3) In current FinFET fabrication, after shallow trench isolation (STI) is formed around the fin-shaped structure part of the fin-shaped structure and part of the STI could be removed to form a trench, and insulating material is deposited into the trench to form single diffusion break (SDB) structure or isolation structure. However, the integration of the SDB structure and metal gate fabrication still remains numerous problems such as window limitation in lithography process. Hence how to improve the current FinFET fabrication to resolve this issue has become an important task in this field.

SUMMARY OF THE INVENTION

(4) According to an embodiment of the present invention, a method for fabricating minimal fin length includes the steps of first forming a fin-shaped structure extending along a first direction on

a substrate, forming a first single-diffusion break (SDB) trench and a second SDB trench extending along a second direction to divide the fin-shaped structure into a first portion, a second portion, and a third portion, and then performing a fin-cut process to remove the first portion and the third portion.

(5) According to another aspect of the present invention, a semiconductor device includes a fin-shaped structure extending along a first direction on a substrate, a first single diffusion break (SDB) structure extending along a second direction on one side of the fin-shaped structure, and a second SDB structure extending along the second direction on another side of the fin-shaped structure. Preferably, the first SDB structure includes a first side contacting the fin-shaped structure and a second side not contacting any fin-shaped structure.

(6) These and other objectives of the present invention will no doubt become obvious to those of ordinary skill in the art after reading the following detailed description of the preferred embodiment that is illustrated in the various figures and drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIGS. **1-6** illustrate a method for fabricating a semiconductor device according to an embodiment of the present invention.

(2) FIG. **7** illustrates a structural view of a semiconductor device according to an embodiment of the present invention.

(3) FIGS. **8-10** illustrate a method for fabricating a semiconductor device according to an embodiment of the present invention.

DETAILED DESCRIPTION

(4) Referring to FIGS. **1-6**, FIGS. **1-6** illustrate a method for fabricating a semiconductor device according to an embodiment of the present invention, in which the bottom left portions of FIGS. **1-6** illustrate cross-sectional views of the top portions of FIGS. **1-6** for fabricating the semiconductor device along the sectional line AA' and bottom right portions of FIGS. **1-6** illustrate cross-sectional views of the top portions of FIGS. **1-6** for fabricating the semiconductor device along the sectional line BB'. As shown in FIG. **1**, a substrate **12**, such as a silicon substrate or silicon-on-insulator (SOI) substrate is first provided. Next, a dummy region **14** and an active region **16** are defined on the substrate **12**, and a plurality of fin-shaped structures **18** are formed on the substrate **12** and extending from the dummy region **14** to the active region **16**. Preferably, at least a dummy fin-shaped structure having minimal fin length and dummy gate structures standing astride the dummy fin-shaped structure will be formed on the dummy region **14** in the later process whereas fin-shaped structures with normal fin length and active gate structures standing astride the fin-shaped structures will be formed on the active region **16** afterwards. It should be noted that even though ten fin-shaped structures **18** are disposed on the substrate **12** in this embodiment, it would also be desirable to adjust the number of fin-shaped structures **18** depending on the demand of the product, which is also within the scope of the present invention.

(5) Preferably, the fin-shaped structures **18** of this embodiment could be obtained by a sidewall image transfer (SIT) process. For instance, a layout pattern is first input into a computer system and is modified through suitable calculation. The modified layout is then defined in a mask and further transferred to a layer of sacrificial layer on a substrate through a photolithographic and an etching process. In this way, several sacrificial layers distributed with a same spacing and of a same width are formed on a substrate. Each of the sacrificial layers may be stripe-shaped. Subsequently, a deposition process and an etching process are carried out such that spacers are formed on the sidewalls of the patterned sacrificial layers. In a next step, sacrificial layers can be removed completely by performing an etching process. Through the etching process, the pattern defined by

the spacers can be transferred into the substrate underneath, and through additional fin cut processes, desirable pattern structures, such as stripe patterned fin-shaped structures could be obtained.

(6) Alternatively, the fin-shaped structures **18** could also be obtained by first forming a patterned mask (not shown) on the substrate, **12**, and through an etching process, the pattern of the patterned mask is transferred to the substrate **12** to form the fin-shaped structures **18**. Moreover, the formation of the fin-shaped structures **18** could also be accomplished by first forming a patterned hard mask (not shown) on the substrate **12**, and a semiconductor layer composed of silicon germanium is grown from the substrate **12** through exposed patterned hard mask via selective epitaxial growth process to form the corresponding fin-shaped structures **18**. These approaches for forming fin-shaped structure are all within the scope of the present invention.

(7) Next, as shown in FIG. 2, a photo-etching process or fin remove process is conducted to selectively remove part of the fin-shaped structures **18** outside the dummy region **14** and active region **16**, such as the fin-shaped structures **18** immediately adjacent to the top edge and bottom edge of the dummy region **14** as shown in the top portion of FIG. 2. As shown in the bottom left portion of FIG. 2, after removing part of the fin-shaped structures **18** above and below top edge and bottom edge of the dummy region **14** the remaining fin-shaped structures **18** preferably form bumps **20** on the substrate **12**, in which the height of the bumps **20** could lower than half or even one third of the total height of the original fin-shaped structures **18** adjacent to two sides of the bumps **20**, which are all within the scope of the present invention.

(8) Next, as shown in FIG. 3, a single diffusion break (SDB) structure such as a first SDB structure **22** and a second SDB structure **24** are formed adjacent to two sides of the dummy region **14** crossing the fin-shaped structures **18**. In this embodiment, the formation of the first SDB structure **22** and second SDB structure **24** could be accomplished by first forming a patterned mask (not shown) on the substrate **12**, and then conducting an etching process by using the patterned mask as mask to remove part of the fin-shaped structure **18** adjacent to two sides of the dummy region **14** along a vertical direction (or Y-direction) to form trenches **26** or SDB trenches and at the same time divide the fin-shaped structures **18** into three portions. Specifically, the trenches **26** formed adjacent to two sides of the dummy region **14** divide each of the fin-shaped structures **18** into three portions, including a first portion **28** on left side of the first SDB structure **22**, a third portion **32** on right side of the second SDB structure **24**, and a second portion **30** between the first SDB structure **22** and the second SDB structure **24**, in which the trenches **26** are formed to defined the position of the SDB structures formed afterwards.

(9) Next, a dielectric layer **34** is formed to fill each of the trenches **26** completely, a planarizing process such as an etching back process or chemical mechanical polishing (CMP) process is conducted to remove part of the dielectric layer **34**, and a photo-etching process is conducted to remove part of the dielectric layer **34** outside the trenches **26** so that the top surface of the remaining dielectric layer **34** is slightly lower than the top surface of the fin-shaped structures **18** for forming a first SDB structure **22** and a second SDB structure **24**. In this embodiment, each of the fin-shaped structures **18** is disposed extending along a first direction (such as X-direction) while the first SDB structure **22** and second SDB structure **24** adjacent to two sides of the dummy region **14** are disposed extending along a second direction (such as Y-direction) orthogonal to the first direction.

(10) Next, as shown in FIG. 4, a fin cut process is conducted to remove the first portions **28** and third portions **32** of the aforementioned fin-shaped structures **18**. In this embodiment, the fin cut process could be accomplished by first forming a patterned mask **36** such as a patterned resist on the substrate **12**, in which the patterned mask **36** preferably covers part of the first SDB structure **22**, the second portion **30** on the dummy region **14**, part of the second SDB structure **24**, and the active region **16** adjacent to the dummy region **14** while other regions not covered by the patterned mask **36** are exposed.

(11) Next, as shown in FIG. 5, an etching process is conducted by using the patterned mask **36** as mask to remove all of the fin-shaped structures **18** outside the first SDB structure **22**, the second portions **30**, the second SDB structure **24**, and the active region **16** such as the first portions **28** and the third portions **32**. After removing the patterned mask **36**, a shallow trench isolation (STI) **38** is formed around the dummy region **14** and the active region **16**. In this embodiment, each of the SDB structures and the STI **38** are formed at different stages while the two elements such as the SDB structures and the STI **38** could be made of same material or different materials depending on the demand of the product. For instance, the first SDB structure **22** and the second SDB structure **24** are preferably made of silicon nitride (SiN) whereas the STI **38** is made of silicon oxide in this embodiment, but not limited thereto.

(12) Next, as shown in FIG. 6, gate structures **40**, **42**, **44**, **46**, **48** are formed on the fin-shaped structures **18** on the dummy region **14** and active region **16**. In this embodiment, the formation of the gate structures **40**, **42**, **44**, **46**, **48** could be accomplished by a gate first process, a high-k first approach from gate last process, or a high-k last approach from gate last process. Since this embodiment pertains to a high-k last approach, a gate dielectric layer **50** or interfacial layer, a gate material layer **52** made of polysilicon, and at least a selective hard mask (not shown) could be formed sequentially on the substrate **12**, and a photo-etching process is then conducted by using a patterned resist (not shown) as mask to remove part of the gate material layer **52** and part of the gate dielectric layer **50** through single or multiple etching processes. After stripping the patterned resist, gate structures **40**, **42**, **44**, **46**, **48** each composed of a patterned gate dielectric layer **50** and a patterned gate material layer **52** are formed on the substrate **12**.

(13) Next, at least a spacer **54** is formed on the sidewalls of the each of the gate structures **40**, **42**, **44**, **46**, **48** and a doped region **56** and/or source/drain regions **58** could be formed in the fin-shaped structures **18** adjacent to two sides of the spacers **54**. For instance, a doped region **56** could be formed between the gate structures **40**, **42** on the dummy region **14** while source/drain regions **58** could be formed adjacent to two sides of the gate structures **46** on the active region **16**. In this embodiment, each of the spacers **54** could be a single spacer or a composite spacer, such as a spacer including but not limited to for example an offset spacer and a main spacer. Preferably, the offset spacer and the main spacer could include same material or different material while both the offset spacer and the main spacer could be made of material including but not limited to for example SiO₂, SiN, SiON, SiCN, or combination thereof. The source/drain regions **58** could include n-type dopants or p-type dopants depending on the type of device being fabricated. Moreover, the source/drain regions **58** could also include different epitaxial materials depending on the type of transistor being fabricated. For instance, the source/drain regions **58** on NMOS region could include silicon carbide (SiC) or silicon phosphide (SiP) whereas source/drain regions **58** on PMOS region could include silicon germanium (SiGe).

(14) Next, an interlayer dielectric (ILD) layer (not shown) could be formed on the gate structures **40**, **42**, **44**, **46**, **48** and a replacement metal gate (RMG) process could be conducted to transform the gate structures **40**, **42**, **44**, **46**, **48** into metal gates. Since the transformation of gate structures **40**, **42**, **44**, **46**, **48** into metal gates are well known to those skilled in the art, the details of which are not explained herein for the sake of brevity. It should be noted that the second portion **30** of the fin-shaped structures on the dummy region **14** formed through the aforementioned process are in fact fin-shaped structures having minimal lengths so that gate structures **40**, **42** could only be formed immediately adjacent to front end and back end of the second portions **30** to form dummy gates due to insufficient length of the fin-shaped structures whereas operational gate structure **46** with source/drain regions **58** adjacent to two sides of the gate structure **46** could be formed on fin-shaped structures **18** having sufficient length for forming operations transistors.

(15) Referring to FIG. 6, FIG. 6 further illustrates a structural view of a semiconductor device according to an embodiment of the present invention. As shown in FIG. 6, the semiconductor device includes a dummy region **14** and an active region **16** defined on the substrate **12**, a plurality

of fin-shaped structures **18** extending along a first direction such as X-direction on the dummy region **14** and active region **16**, a first SDB structure **22** extending along a second direction such as Y-direction on one side such as left side of the fin-shaped structures **18** on the dummy region **14**, a second SDB structure **24** extending along the same second direction on another side such as right side of the fin-shaped structures **18** on the dummy region **14**, a gate structure **40** extending along the Y-direction and overlapping one side such as left side of the fin-shaped structures **18** on the dummy region **14**, a gate structure **42** extending along the Y-direction and overlapping another side such as right side of the fin-shaped structures **18** on the dummy region **14**, a gate structure **44** extending along the Y-direction on one side such as left side of the fin-shaped structures **18** on the active region **16**, a gate structure **48** extending along the Y-direction on another side such as right side of the fin-shaped structures **18** on the active region **16**, and a gate structure **46** extending along the Y-direction between the gate structures **44**, **48** on the active region **16**.

(16) Viewing from a more detailed perspective, an edge such as left edge or left sidewall of the gate structure **40** is aligned with left edge of the fin-shaped structures **18** on the dummy region **14**, an edge such as right edge or right sidewall of the gate structure **42** is aligned with right edge of the fin-shaped structures **18** on the dummy region **14**, the first SDB structure **22** includes a first side such as right side directly contacting the fin-shaped structure **18** and a second side such as left side not contacting any of the fin-shaped structures directly. Similarly, the second SDB structure **24** includes a third side such as left side directly contacting the fin-shaped structures **18** and a fourth side such as right side not contacting any of the fin-shaped structures directly.

(17) As illustrate by the cross-section view shown in bottom right corner of FIG. **6**, the bottom surface of the first SDB structure **22** on left side of the dummy region **14** and the bottom surface of the second SDB structure **24** on right side of the dummy region **14** are slightly lower than the top surface of the substrate **12** or viewing from another angle, the top surface of the substrate **12** directly under the first SDB structure **22** is slightly lower than the top surface of the substrate **12** directly under the fin-shaped structures **18** on the dummy region **14**. Similarly, the top surface of the substrate **12** directly under the second SDB structure **24** is slightly lower than the top surface of the substrate **12** directly under the fin-shaped structures **18** on the dummy region **14**.

(18) Referring to FIG. **7**, FIG. **7** further illustrates a structural view of a semiconductor device according to an embodiment of the present invention. As shown in bottom right corner of FIG. **7**, in contrast to the aforementioned embodiment of using fin cut process to completely remove the first portions **28** and third portions **32** so that the left side of the first SDB structure **22** not contacting any of the fin-shaped structures directly and the right side of the second SDB structure **24** not contacting any of the fin-shaped structures directly, according to other embodiment of the present invention it would also be desirable to conduct the fin cut process and remain part of the fin-shaped structures **18** to form bumps **60** during removal of the first portions **28** and third portions **32** so that right side of the first SDB structure **22** would contact the fin-shaped structures **18** while left side of the first SDB structure **22** would contact the bump **60** directly and left side of the second SDB structure **24** would contact the fin-shaped structures **18** while right side of the second SDB structure **24** would contact the bump **60** directly, which is also within the scope of the present invention.

(19) Referring to FIGS. **8-10**, FIGS. **8-10** illustrate a method for fabricating a semiconductor device according to an embodiment of the present invention. As shown in FIG. **8**, instead of forming trenches **26** adjacent to two sides of the dummy region **14**, forming a dielectric layer **34** in the trenches **26** to form the first SDB structure **22** and second SDB structure **24**, and then conducting a fin cut process as conducted in FIGS. **3-5**, it would also be desirable to conduct an etching process by using a patterned mask (not shown) to remove part of the fin-shaped structures **18** adjacent to two sides of the dummy region **14** along the extending direction of the fin-shaped structures **18** for forming trenches **26** and at the same time divide the fin-shaped structures **18** into three portions, in which the trenches **26** could also be referred to as SDB trenches.

(20) Next, as shown in FIG. 9, the aforementioned fin cut process conducted in FIGS. 4-5 could be carried out after the trenches 26 are formed by using the patterned mask 36 to remove the first portions 28 and third portions 32 of the fin-shaped structures 18 outside the dummy region 14 and active region 16, and the patterned mask 36 is removed thereafter.

(21) Next, as shown in FIG. 10, a dielectric layer 34 is formed to fill the trenches 26 and area outside the trenches 26 completely, and an etching back process is conducted to remove part of the dielectric layer 34 to form a first SDB structure 22 and second SDB structure 24 in the trenches 26 and at the same time form a STI 38 adjacent to the first SDB structure 22 and second SDB structure 24. Since the first SDB structure 22, the second SDB structure 24, and the STI 38 are formed at the same time at this stage, the three elements 22, 24, 38 are preferably made of same material such as silicon oxide. Next, the process conducted in FIG. 6 could be carried out to form gate structures on the fin-shaped structures 18 on the dummy region 14 and active region 16, and the details of which are not explained herein for the sake of brevity.

(22) Overall, the present invention discloses an approach for fabricating fin-shaped structures having minimal length, in which a first SDB structure and a second SDB structure are first formed on the dummy region 14 to divide fin-shaped structures into three portions, a patterned mask is formed on the first SDB structure, the second SDB structure, and a second portion of fin-shaped structures between the two SDB structures, and then a fin cut process is conducted to remove the first portion and third portion of the fin-shaped structures adjacent to two sides of the SDB structures. In contrast to the conventional approach of using patterned to conduct fin cut process and unable to accurately focus and remove fin-shaped structures outside the dummy region and active region due to limitation of lithography process, the present invention preferably uses two SDB structures for expanding the window of photolithography process thereby obtaining better focus and alignment. Moreover, the second portions 30 of fin-shaped structures formed on the dummy region 14 through aforementioned process are in fact fin-shaped structures having minimal length such that it would only be possible to form two dummy gate structures 40, 42 on front and back end of the fin-shaped structures respectively while an operational gate structure 46 with accompanying source/drain regions 58 on two sides could be formed on fin-shaped structures having standard or regular length on the active region 16.

(23) Those skilled in the art will readily observe that numerous modifications and alterations of the device and method may be made while retaining the teachings of the invention. Accordingly, the above disclosure should be construed as limited only by the metes and bounds of the appended claims.

Claims

1. A semiconductor device, comprising: a fin-shaped structure extending along a first direction on a substrate; a first single diffusion break (SDB) structure extending along a second direction on one side of the fin-shaped structure, wherein the first SDB structure comprises a first side contacting the fin-shaped structure and a second side not contacting any fin-shaped structure; a second SDB structure extending along the second direction on another side of the fin-shaped structure; and a first gate structure extending along the second direction on one side of the fin-shaped structure, wherein the first SDB structure is adjacent to the first gate structure and not directly under the first gate structure and sidewalls of the first gate structure and the first SDB structure are aligned.
2. The semiconductor device of claim 1, wherein the second SDB structure comprises a third side contacting the fin-shaped structure and a fourth side not contacting any fin-shaped structure.
3. The semiconductor device of claim 1, comprising: a second gate structure extending along the second direction on another side of the fin-shaped structure.
4. The semiconductor device of claim 3, wherein an edge of the first gate structure is aligned with an edge of the fin-shaped structure.

5. The semiconductor device of claim 3, wherein an edge of the second gate structure is aligned with an edge of the fin-shaped structure.
 6. The semiconductor device of claim 1, wherein a top surface of the substrate directly under the first SDB structure is lower than a top surface of the substrate directly under the fin-shaped structure.
 7. The semiconductor device of claim 1, wherein a top surface of the substrate directly under the second SDB structure is lower than a top surface of the substrate directly under the fin-shaped structure.
 8. The semiconductor device of claim 1, wherein the first direction is orthogonal to the second direction.
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